

Example:

- **planet 0 = 100 images**
- **planet 1 = 200 images**
- **planet 2 = 50 images**

And inside **planet 0**, there are 3 sub-planets:

- **sub-planet 0 = 30 images**
- **sub-planet 1 = 60 images**
- **sub-planet 2 = 10 images**

step by step how the size is calculated.

1. Formula used for size

Step 1: Base radius

$$\text{base radius} = \log_2(\text{size} + 1) \times 0.3$$

Step 2: Final radius after depth reduction

$$\text{final radius} = \text{base radius} \times 0.65^{\text{depth}}$$

And if it becomes too small, there is a minimum floor.

2. First calculate the top-level planets

Top-level planets are at **depth 0**.

So for top-level planets:

$$0.65^0 = 1$$

That means:

- for planets, **final radius = base radius**
- because depth 0 does not reduce anything

Planet 0 = 100 images

Step 1

$$size + 1 = 100 + 1 = 101$$

Step 2

$$\log_2(101) \approx 6.66$$

Step 3

$$6.66 \times 0.3 = 1.998$$

So base radius $\approx$ **2.00**

Step 4: depth reduction

planet 0 is top-level, so depth = 0

$$2.00 \times 0.65^0 = 2.00 \times 1 = 2.00$$

Final radius of planet 0

2.00

Planet 1 = 200 images

Step 1

$$200 + 1 = 201$$

Step 2

$$\log_2(201) \approx 7.65$$

Step 3

$$7.65 \times 0.3 = 2.295$$

Base radius ≈ 2.30

Step 4: depth reduction

$$\text{depth} = 0$$

$$2.30 \times 1 = 2.30$$

Final radius of planet 1

2.30

Planet 2 = 50 images

Step 1

$$50 + 1 = 51$$

Step 2

$$\log_2(51) \approx 5.67$$

Step 3

$$5.67 \times 0.3 = 1.701$$

Base radius ≈ 1.70

Step 4: depth reduction

$$\text{depth} = 0$$

$$1.70 \times 1 = 1.70$$

Final radius of planet 2

1.70

3. So after pressing Play, what sizes will the 3 planets have?

Planet	Images	Base radius	Depth	Final radius
planet 0	100	2.00	0	2.00
planet 1	200	2.30	0	2.30
planet 2	50	1.70	0	1.70

So visually:

- **planet 1** will be the biggest
- **planet 0** will be medium
- **planet 2** will be the smallest

Because $200 > 100 > 50$

4. Now click planet 0

You said planet 0 has 3 sub-planets:

- sub-planet 0 = 30 images
- sub-planet 1 = 60 images
- sub-planet 2 = 10 images

These are **children of planet 0**, so they are one level below it.

That means:

- their **depth = 1**

So now, after computing base radius, we must multiply by:

$$0.65^1 = 0.65$$

Sub-planet 0 = 30 images

Step 1

$$30 + 1 = 31$$

Step 2

$$\log_2(31) \approx 4.95$$

Step 3

$$4.95 \times 0.3 = 1.485$$

Base radius ≈ 1.49

Step 4: apply depth 1

$$1.49 \times 0.65 = 0.9685$$

Final radius of sub-planet 0

0.97

Sub-planet 1 = 60 images

Step 1

$$60 + 1 = 61$$

Step 2

$$\log_2(61) \approx 5.93$$

Step 3

$$5.93 \times 0.3 = 1.779$$

Base radius ≈ 1.78

Step 4: apply depth 1

$$1.78 \times 0.65 = 1.157$$

Final radius of sub-planet 1

1.16

Sub-planet 2 = 10 images

Step 1

$$10 + 1 = 11$$

Step 2

$$\log_2(11) \approx 3.46$$

Step 3

$$3.46 \times 0.3 = 1.038$$

Base radius $\approx$ **1.04**

Step 4: apply depth 1

$$1.04 \times 0.65 = 0.676$$

Final radius of sub-planet 2

0.68

5. So after clicking planet 0, what sizes will the 3 sub-planets have?

Sub-planet	Images	Base radius	Depth	Final radius
sub-planet 0	30	1.49	1	0.97
sub-planet 1	60	1.78	1	1.16
sub-planet 2	10	1.04	1	0.68

So visually:

- **sub-planet 1** will be the biggest among the 3

- **sub-planet 0** will be medium
- **sub-planet 2** will be the smallest

Because $60 > 30 > 10$

6. Compare planet 0 and its sub-planets

Node	Images	Final radius
planet 0	100	2.00
sub-planet 0	30	0.97
sub-planet 1	60	1.16
sub-planet 2	10	0.68

This shows two things:

First

Bigger cluster → bigger sphere

Second

Deeper level → smaller sphere

That is why sub-planets are smaller than the original planet.

7. Why do we use this logic?

Because if we used raw image count directly:

- 200 images → radius 200
- 60 images → radius 60
- 10 images → radius 10

then the planets would become absurdly huge.

So instead we do:

- **logarithm** to compress the size
- **depth reduction** to make deeper levels smaller

8. Very short version of your example

Top-level planets

- planet 0 (100) → **2.00**
- planet 1 (200) → **2.30**
- planet 2 (50) → **1.70**

Children of planet 0

- sub-planet 0 (30) → **0.97**

- sub-planet 1 (60) → **1.16**
- sub-planet 2 (10) → **0.68**

Note:

1. First minimum floor:

`minRadius = 0.3`

This is applied when calculating the **base radius**.

$$\text{baseRadius} = \max(\log_2(\text{size} + 1) \times 0.3, 0.3)$$

This means:

- first calculate $\log_2(\text{size} + 1) \times 0.3$
- if that result is **smaller than 0.3**
- then use **0.3 instead**

So the base radius can never go below **0.3**

2. Second minimum floor:

`minEffectiveRadius = 0.1`

This is applied **after depth reduction**.

$$\text{effectiveRadius} = \max(\text{baseRadius} \times 0.65^{\text{depth}}, 0.1)$$

This means:

- take the base radius
- shrink it by depth
- if the result becomes **smaller than 0.1**
- then use **0.1 instead**

So the final visible sphere radius can never go below **0.1**